Electronic Supplementary Information (ESI): Multi-Physics Derivations and Bounding Analysis

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S1. Buckingham Pi Scaling Invariance

System mass scaling across five orders of magnitude (25 kg to $1,000,000$ kg) is dictated by the fundamental dimensionless invariant:

$$\Pi_1 = \lambda \cdot S_m^{-1/3} = 1.0000 \tag{1}$$

where λ represents the linear geometric dimension scale factor and $S_m = M/M_{\text{ref}}$ is the mass ratio relative to the $20,000$ kg Tier 3 platform.

S2. AC MHD Traveling Wave Vector Potential

The electrodeless AC MHD propulsion channels generate a traveling wave magnetic field via RF induction coils at $f_{\text{RF}} = 25.0$ kHz:

$$\mathbf{B}(x,t) = B_0 \sin(\omega t - kx)\hat{\mathbf{y}} \tag{2}$$

The electromagnetic skin depth in seawater ($\sigma = 4.5$ S/m) is given by:

$$\delta = \sqrt{\frac{1}{\pi f_{\text{RF}} \mu_0 \sigma}} = 0.15 \text{ m} \tag{3}$$

S3. Anisotropic Tsai-Wu Tensor Failure Proof

Under $4,000$ m hydrostatic compression, the principal stress state in Quadrant III is $(\sigma_1, \sigma_2) = (-158.5, -110.0)$ MPa. The Tsai-Wu tensor metric evaluates as:

$$f_{\text{TW}} = F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + 2F_{12}\sigma_1\sigma_2 = 0.244 \leq 1.0 \tag{4}$$

yielding a structural Factor of Safety FOS $= 1/f_{\text{TW}} = 4.10$.

S4. ISS Lyapunov Control Eigenvalue Analysis

Non-linear transition stability across seawater-air interface piercing is governed by the Lyapunov equation $A^T P + PA = -Q$. The eigenvalues of P are strictly positive real:

$$\lambda(P) \in \{1.43, 2.06, 3.31\} \tag{5}$$

proving Input-to-State Stability (ISS).